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Solution: B

L3

To simulate gravity

→ normal contact force $N = mg$; where g = acceleration due to gravity

Centripetal force is provided by the normal contact force

$$F_c = N$$

$$F_c = ma_c = mr\omega^2 = mg$$

$$r\omega^2 = g$$

 $g \propto r$ since ω^2 is a constant

$$\frac{9.81}{2150} = \frac{3.72}{r}$$
$$r = 815 \text{ m}$$

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Solution: C